

Differential Cryptanalysis of Reduced-Round PRESENT*

Meiqin Wang

Key Laboratory of Cryptologic Technology and Information Security, Ministry of
Education, Shandong University,
Jinan, 250100, China
mqwang@sdu.edu.cn

Abstract. PRESENT is proposed by A.Bogdanov et al. in CHES 2007 for extremely constrained environments such as RFID tags and sensor networks. In this paper, we present the differential characteristics for r -round ($5 \leq r \leq 15$), then give the differential cryptanalysis on reduced-round variants of PRESENT. We attack 16-round PRESENT using 2^{64} chosen plaintexts, 2^{32} 6-bit counters, and 2^{64} memory accesses.

1 Introduction

RFID systems and sensor networks have been aggressively deployed in a variety of applications, but their further pervasive usage is mainly limited by lots of security and privacy concerns. As RFID tags and sensor networks are low cost with limited resources, the present cryptographic primitives can not be feasible. So the security primitives suitable for these environments must be designed.

PRESENT is an Ultra-Lightweight block cipher proposed by A.Bogdanov, L.R.Knudsen and G.Leander et al.[3] and has implementation requirements similar to many compact stream ciphers. Compared to other current block ciphers for low-cost implementation requirements such as TEA[12,13], MCRYPTON[7], HIGHT[5], SEA[11] and CGEN[9], PRESENT has the lowest implementation costs.

PRESENT is a 31-round SP-network with block length 64 bits and 80 bits or 128 bits key length. Serpent[1] and DES have excellent performance in hardware, so the design of PRESENT makes use of the characteristics of the two block ciphers. The non-linear substitution layer S-box of PRESENT is similar to that of Serpent and the linear permutation layer pLayer of PRESENT is similar to that of DES.

Differential cryptanalysis, proposed by Biham and Shamir[4], is one of the most general cryptanalytic techniques. Although the original PRESENT proposal provided theoretical upper bounds for the highest probability characteristics of 25-round PRESENT[3], the proposal did not give the concrete differential cryptanalysis.

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In this paper, we consider actual differential attack against reduced-round PRESENT. First we give some differential characteristics for PRESENT. 14-round differential characteristics occur with the probability of 2^{-62} and 15-round differential characteristics occur with the probability of 2^{-66} . However, the original PRESENT proposal can provided theoretical upper bounds for the characteristics of 15-round PRESENT with the highest probability 2^{-60} . Second, we attack 16-round PRESENT with 14-round differential characteristics using 2^{64} chosen plaintexts, 2^{32} 6-bit counters, and 2^{64} memory accesses.

The paper is organized as follows. Section 2 introduces the description of PRESENT. In Section 3, we give some notations used in this paper. In Section 4, we present the best differential characteristics we found for PRESENT, and give the differential attack on 16-round PRESENT-80. Section 5 concludes this paper.

2 Description of PRESENT

2.1 The Encryption Process

PRESENT is a 31-round Ultra-Lightweight block cipher. The block length is 64-bit. PRESENT uses only one 4-bit S-box S which is applied 16 times in parallel in each round. The cipher is described in Figure 1. As in Serpent, there are three stages involved in PRESENT. The first stage is addRoundKey described as follows,

$$b_j \rightarrow b_j \oplus k_j^i$$

where $b_j, 0 \leq j \leq 63$ is the current state and $k_j^i, 1 \leq i \leq 32, 0 \leq j \leq 63$ is the j -th subkey bit of round key K_i .

The second stage is sBoxLayer which consists of 16 parallel versions of the 4-bit to 4-bit S-box, which is given in Table 1.

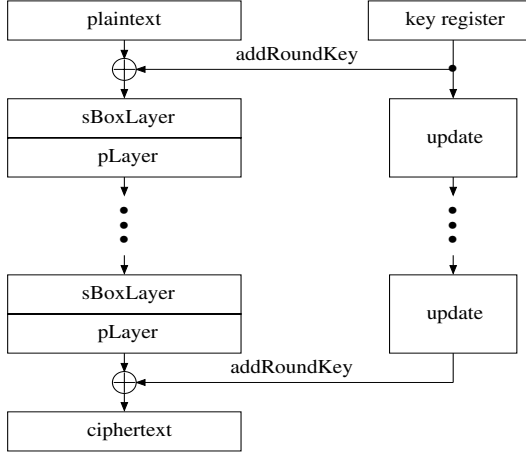
Table 1. Table of S-box

x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
S[x]	C	5	6	B	9	0	A	D	3	E	F	8	4	7	1	2

The third stage is the bit permutation pLayer, which is given by Table 2. From pLayer, bit i of stage is moved to bit position $P(i)$.

2.2 The Key Schedule

PRESENT's key schedule can take key sizes of 80 bits or 128 bits. We will cryptanalyze 80 bits version, so we will only give the schedule algorithm for 80 bits version.

**Fig. 1.** 31-round PRESENT Encryption Algorithm**Table 2.** Table of pLayer

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$P(i)$	0	16	32	48	1	17	33	49	2	18	34	50	3	19	35	51
i	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
$P(i)$	4	20	36	52	5	21	37	53	6	22	38	54	7	23	39	55
i	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
$P(i)$	8	24	40	56	9	25	41	57	10	26	42	58	11	27	43	59
i	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
$P(i)$	12	28	44	60	13	29	45	61	14	30	46	62	15	31	47	63

Firstly, the 80-bit key will be stored in a key register K denoted as $K = k_{79}k_{78} \dots k_0$. In round j , PRESENT firstly extracts 64-bit subkeys K_j in the following ways,

$$K_j = \kappa_{63}\kappa_{62} \dots \kappa_0 = k_{79}k_{78} \dots k_{16}$$

Then it updates key register $K = k_{79}k_{78} \dots k_0$ as follows,

$$[k_{79}k_{78} \dots k_1k_0] = [k_{18}k_{17} \dots k_{20}k_{19}]$$

$$[k_{79}k_{78}k_{77}k_{76}] = S[k_{79}k_{78}k_{77}k_{76}]$$

$$[k_{19}k_{18}k_{17}k_{16}k_{15}] = [k_{19}k_{18}k_{17}k_{16}k_{15}] \oplus round_counter$$

3 Some Notations

In the remainder of the paper, we use $X = x_0, x_1, \dots, x_{15}$ to denote the intermediate difference in each step. $x_0, x_1, \dots, x_{15}$ are 16 nibble differences and x_0 is

2-round to 4-round are 4, 7 and 9 respectively. The possible distribution of the number of active S-boxes in them is listed in Table 4. As a result, only 4-round iterative characteristics with the probability 2^{-18} have been found, one of which is given in Table 5.

- We searched the best differential characteristics from 5-round to 10-round which are more advantage than the characteristics based on 4-round iterative characteristics we have found.
- Based on 4-round iterative characteristic, we have found the differential characteristics from 11-round to 15-round.

Table 4. Possible Distribution of Active S-box for Iterative Characteristics

Rounds	2	3	4
Possible Distribution of Active S-box	2-2	2-2-2	2-2-2-2
		3-2-2	3-2-2-2
		2-3-2	2-3-2-2
		2-2-3	2-2-3-2
			2-2-2-3

Table 5. 4-round Iterative Differential of PRESENT

Rounds		Differences	Pr
I		$x_0 = 4, x_3 = 4$	1
R1	S	$x_0 = 5, x_3 = 5$	$\frac{1}{2^4}$
R1	P	$x_0 = 9, x_8 = 9$	1
R2	S	$x_0 = 4, x_8 = 4$	$\frac{1}{2^4}$
R2	P	$x_8 = 1, x_{10} = 1$	1
R3	S	$x_8 = 9, x_{10} = 9$	$\frac{1}{2^4}$
R3	P	$x_2 = 5, x_{14} = 5$	1
R4	S	$x_2 = 1, x_{14} = 1$	$\frac{1}{2^6}$
R4	P	$x_0 = 4, x_3 = 4$	1

The differential characteristics we found in the way are given in Table 6. It is noted that the number of active S-boxes in each round are all 2.

We have found 24 14-round differential characteristics with the probability 2^{-62} with different input differences but the same output difference. All the characteristics have the same differences after the 8 – th round, and all have the same probability 2^{-62} . Table 7 gives one of the 14-round characteristics we have found.

Table 6. Probability of the Best Characteristics We Found

Rounds	Differential Probability	Number of Active S-box
5	2^{-20}	10
6	2^{-24}	12
7	2^{-28}	14
8	2^{-32}	16
9	2^{-36}	18
10	2^{-42}	20
11	2^{-46}	22
12	2^{-52}	24
13	2^{-56}	26
14	2^{-62}	28
15	2^{-66}	30

4.2 Attacking 16-Round PRESENT

We will attack 16-round PRESENT using the 14-round differential characteristics with probability of 2^{-62} .

All of the 24 differential characteristics we found have 2 active S-boxes in the first round located in position 0,1,2,12,13 and 14, so the S-boxes in position from 3 to 11 and 15 are all non-active. This attack requires 2^{40} structures of 2^{24} chosen plaintexts each. In each structure, all the inputs to the 14 non-active S-boxes in the first round can be random value, while 8 bits of input to 2 active S-boxes take 2^8 possible values. In all structures, there are $2^{40} \cdot 2^{16} \cdot 2^7 = 2^{63}$ pairs for each possible characteristics. Each characteristic has the probability 2^{-62} , so the number of right pairs is $2^{63} * 2^{-62} * 24 = 48$ satisfying any one characteristic. For each structure, the number of possible pairs is $(2^{24})^2 / 2 = 2^{47}$, thus we have $2^{47} \cdot 2^{40} = 2^{87}$ pairs of plaintext to be considered in total.

According to the output difference of 14-round differential characteristics, there are two active S-boxes in round-15 which are x_0 and x_8 whose input difference is 9 and output difference will be 2, 4, 6, 8, 12 or 14. The least significant bit of their output difference must be zero, so at most 6 bits are non-zero for the output difference of S-boxes in round 15. After the pLayer of round 15, the maximum number of active S-boxes for round 16 is 6 and the active S-boxes will be $x_4, x_6, x_8, x_{10}, x_{12}$ and x_{14} , and the minimum number of active S-boxes for round 16 is 2.

For each structure, each pair satisfying one of the characteristics should have 10 non-active S-boxes in round 16, so the wrong pairs should be discarded. Thus about $2^{47} * 2^{-40} = 2^7$ candidates for right pairs remain from each structure.

Table 7. The 14-round Differential of PRESENT

Rounds		Differences	Pr
I		$x_2 = 7, x_{14} = 7$	
R1	S	$x_2 = 1, x_{14} = 1$	$\frac{1}{2^4}$
R1	P	$x_0 = 4, x_3 = 4$	1
R2	S	$x_0 = 5, x_3 = 5$	$\frac{1}{2^4}$
R2	P	$x_0 = 9, x_8 = 9$	1
R3	S	$x_0 = 4, x_8 = 4$	$\frac{1}{2^4}$
R3	P	$x_8 = 1, x_{10} = 1$	1
R4	S	$x_8 = 9, x_{10} = 9$	$\frac{1}{2^4}$
R4	P	$x_2 = 5, x_{14} = 5$	1
R5	S	$x_2 = 1, x_{14} = 1$	$\frac{1}{2^6}$
R5	P	$x_0 = 4, x_3 = 4$	1
R6	S	$x_0 = 5, x_3 = 5$	$\frac{1}{2^4}$
R6	P	$x_0 = 9, x_8 = 9$	1
R7	S	$x_0 = 4, x_8 = 4$	$\frac{1}{2^4}$
R7	P	$x_8 = 1, x_{10} = 1$	1
R8	S	$x_8 = 9, x_{10} = 9$	$\frac{1}{2^4}$
R8	P	$x_2 = 5, x_{14} = 5$	1
R9	S	$x_2 = 1, x_{14} = 1$	$\frac{1}{2^6}$
R9	P	$x_0 = 4, x_3 = 4$	1
R10	S	$x_0 = 5, x_3 = 5$	$\frac{1}{2^4}$
R10	P	$x_0 = 9, x_8 = 9$	1
R11	S	$x_0 = 4, x_8 = 4$	$\frac{1}{2^4}$
R11	P	$x_8 = 1, x_{10} = 1$	1
R12	S	$x_8 = 9, x_{10} = 9$	$\frac{1}{2^4}$
R12	P	$x_2 = 5, x_{14} = 5$	1
R13	S	$x_2 = 1, x_{14} = 1$	$\frac{1}{2^6}$
R13	P	$x_0 = 4, x_3 = 4$	1
R14	S	$x_0 = 5, x_3 = 5$	$\frac{1}{2^4}$
R14	P	$x_0 = 9, x_8 = 9$	1

Among 16 S-boxes in round 16, 10 S-boxes must be non-active, 2 S-boxes must be active and 4 S-boxes can be active or non-active. If it is active, the input difference must be 1, and the output difference will be 3, 7, 9 or 13. Discarding any pair with a wrong output difference using the above filter should keep only a fraction of $\frac{5}{16}^6 = 2^{-10.07}$. So only about $2^7 * 2^{-10.07} = 2^{-3.07}$ pairs remain for each structure.

For each structure, we check if the remaining pairs satisfy one of the 24 possible plaintext differences corresponding to 24 characteristics. As there are about 2^{24}

possible input differences, only a fraction of about $2^{-24} * 20 = 2^{-19.68}$ of the pairs remain. So the expected number of remaining pairs in all the 2^{40} structures is $2^{40} * 2^{-3.07} * 2^{-19.68} = 2^{17.25}$.

Only the ciphertext bits corresponding to active S-boxes in the last two round need to be decrypt, so 8 bits of round subkey K_{16} and 24 bits of round subkey K_{17} will be involved during decrypt from round 16 to round 14. After deriving the subkey K_{16} and K_{17} from the master key K , the 24-bit subkey from K_{17} above are independent from the 8-bit subkey from K_{16} , so the total number of subkey bits involved in the decryption from round 16 to round 14 is 32.

For each remaining pair, we guess the 24-bit subkey of K_{17} and 8-bit subkey of K_{16} in round 16, and decrypt the remaining ciphertext pairs from round 16 to round 14. According to the differential distribution table of S-box for PRESENT, given the input difference and output difference, there will be at most 4 pairs occurrences, so the average count per counted pair of the subkey nibble corresponding to one active S-box will be 4. According to the number of active S-boxes in round 16 denoted as t ($2 \leq t \leq 6$) for the remaining ciphertext pairs, we will consider 5 cases according to the value of t :

- If $t = 2$, only $2^{17.25} * 2^{-16} = 2^{1.25}$ pairs of ciphertext satisfy the condition of 2 active S-boxes, so the total counted times of subkeys are about $2^{1.25} * 4^4 = 2^{9.25}$ for the remaining pairs.
- If $t = 3$, about $2^{17.25} * (2^{-12} - 2^{-16}) = 2^{5.16}$ pairs of ciphertext satisfy the condition of 3 active S-boxes, so the total counted times of subkeys are about $2^{5.16} * 4^5 = 2^{15.16}$ for the remaining pairs.
- If $t = 4$, about $2^{17.25} * (2^{-8} - 2^{-12}) = 2^{9.16}$ pairs of ciphertext satisfy the condition of 4 active S-boxes, so the total counted times of subkeys are about $2^{9.16} * 4^6 = 2^{21.16}$ for the remaining pairs.
- If $t = 5$, about $2^{17.25} * (2^{-4} - 2^{-8}) = 2^{13.16}$ pairs of ciphertext satisfy the condition of 5 active S-boxes, so the total counted times of subkeys are about $2^{13.16} * 4^7 = 2^{27.16}$ for the remaining pairs.
- If $t = 6$, the remained pairs satisfying the condition of 6 active S-box will be $2^{17.25} * (1 - 2^{-4}) = 2^{17.16}$, so the total counted times of subkeys are about $2^{17.16} * 4^8 = 2^{33.16}$ for the remaining pairs.

The total counted times of the subkeys are $2^{33.16} + 2^{27.16} + 2^{21.16} + 2^{15.16} + 2^{9.25} = 2^{33.18}$, so the wrong subkey hits are average about $2^{33.18}/2^{32} = 2^{1.18} = 2.27$ times, but the right subkey is counted for the right pairs about 48 times, so it can be easily identified. In total, we retrieve 32 subkey bits using at most $2^{33.18}$ 2-round PRESENT encryptions and 2^{32} 6-bit counters.

By exhaustively searching the remaining 48 bits master key, we can find out 80-bit master key. In this step, the time complexity is 2^{48} 16-round PRESENT encryptions.

In order to reduce the time of analysis we perform the follow algorithm [11]:

1. For each structure:
 - (a) Insert all the ciphertext into the hash table according to the 40-bit ciphertext'bit of the non-active S-boxes in the last round.

- (b) For each entry with collision(a pair of ciphertext with equal 40-bit values) check whether the plaintexts'difference(in round 1) is one of the 24 characteristics's input difference.
 - (c) If a pair passes the above test, check whether the difference(in the 24 bits) can be caused by the output difference of the characteristics.
 - (d) For each possible subkey of K_{17} , we decrypt the last round to obtain the output difference of 2 two active S-boxes for round 15, and check whether the difference(in the 8 bits) can be caused by the output difference of the characteristics. If a pair passes the above test, add 1 to the counter related to 24 bits of K_{17} and 8 bits of K_{16} .
2. Collect all the subkeys whose counter has at least 48 hits. With the high probability the correct subkey is in this list.
 3. Exhaustive searching the remaining 48-bit master key, we can obtain the whole 80-bit master key.

4.3 Complexity Estimations

In step (a), the time complexity is 2^{24} memory accesses. In step (b), about 2^7 pairs remain through the filter of step (a), so the time complexity is 2^8 memory accesses. The time complexity of step (c) (d) (e) and step 2 can be ignored for the fewer remaining pairs for each structure. In all, the time in step 1 is 2^{64} memory accesses.

In step 3, the time complexity is about 2^{48} 16-round PRESENT encryptions.

According to the relationship between the memory accesses and the encryption time in [4], 2^{64} memory accesses is the main term in the implementing time, so the time complexity is about 2^{64} memory accesses.

In our attack, the ratio of signal to noise is as follows:

$$S/N = \frac{p * 2^k}{\alpha * \beta} = \frac{2^{-62} * 2^{32}}{2^{33.18-17.25} * 2^{17.25-67.32}} = 17.63$$

The success probability is as follows:

$$Ps = \int_{-\frac{\sqrt{\mu S_N} - \Phi^{-1}(1-2^{-a})}{\sqrt{S_N+1}}}^{\infty} \Phi(x) dx = 0.999999939$$

where $a = 32$ is the number of subkey bits involved in the decryption and μ is the number of right pairs which can be obtained

$$\mu = pN = 2^{-62} * 2^{63} * 24 = 48$$

In all, our attack needs encrypt the whole plaintext space. The time complexity is 2^{64} memory accesses. The memory requirements are about 2^{32} 6-bit counters and 2^{24} cells for hash table. We can obtain the right key with the probability 0.999999939.

5 Summary

In this paper, we give the differential cryptanalysis on reduced-round variants of PRESENT. We attack 16-round PRESENT using 2^{64} chosen plaintexts, 2^{32} 6-bit counters and 2^{24} hash cells, the time complexity in our attack is about 2^{64} memory accesses. Our attack requires the encryption of the whole plaintext space.

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